# Ankit Prajapati

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.71 |
| GOVT. HIGHER SECONDARY SCHOOL | Class XII (Madhya Pradesh Board of Secondary Education) | 2018 | 83.80% |
| ST. JOHN HIGH SCHOOL | Class X (Madhya Pradesh Board of Secondary Education) | 2016 | 91.60% |

## PROJECTS

**M.Tech Thesis: Performance Modeling & Hybrid Parallelization of 2D Solvers |** C, MPI, OpenMP

- Built and validated explicit (Jacobi) and implicit (ADI/Thomas Algorithm) parallel PDE solvers in C on a 32-core AMD EPYC cluster, verified via the Method of Manufactured Solutions.
- Ran strong scaling studies showing ADI reaches ~13x speedup vs. Jacobi's ~5x at 32 processes heavier per cell computation better masks MPI communication latency.
- Found the optimal MPI rank/OpenMP thread topology is grid size dependent: an 8×4 hybrid split wins at smaller grids, shifting toward pure MPI as the grid grows past L3 cache limits.

**CarbonLedger: AI Carbon Accounting Agent** | *Python, FastAPI, scikit-learn, Gemini API, RAG*

- Built an agent pipeline that turns unstructured invoices into ESRS/CSRD aligned carbon emission reports.
- Cut LLM API calls by ~90% with a TF IDF MLP classifier and confidence gated RAG escalation to Gemini.
- Designed a deterministic Scope 1/2/3 engine, separate from the LLM, for auditable figures and PDF reports.
- Added full audit trail logging and automated PDF report generation (ReportLab) behind a Streamlit interface.

**WealthPilot: AI Agent for SME Loan Underwriting & Credit Research** | Python, LangChain, LangGraph, RAG, Chroma, MCP, Pydantic, pytest

- Built a multi agent LangGraph system for SME credit underwriting with a deterministic ratio engine, policy grounded RAG, persistent case memory, and a human in the loop approval gate with a proven fairness safeguard
- Exposed tools via an MCP server for cross client interoperability, validated with 65 automated tests and a 7x cold start latency improvement (7.6s to 1.1s) through dependency profiling.

**Urban Heat Island Analysis, Hyderabad** | *Python, QGIS, GDAL, Landsat 8, Random Forest*

- Analyzed 7 years of Landsat 8 imagery across 150 wards to quantify spatial and temporal UHI patterns.
- Built a QGIS/GDAL + Pandas pipeline and a Random Forest model linking land use change to surface temperature.
- Measured city wide minimum LST rising from 21.4°C (2016) to 29.6°C (2022), a +8.2°C shift in cooler zones.
- Identified bare land to urban conversion as the strongest factor (Pearson $r = 0.81$, $p < 0.001$).

**Asbestos Fiber Detection -- Object Detection with YOLOv8 |** Python, PyTorch, Ultralytics YOLOv8, Gradio

- Fine-tuned a YOLOv8 object detector via transfer learning on 1,300+ real microscopic images across 4 fiber classes, achieving 0.61 mAP@0.5 and 0.72 precision on held-out validation data.
- Diagnosed class-specific failure modes through confusion-matrix and qualitative error analysis, identifying a visual difficulty pattern independent of training data volume.
- Built and deployed an interactive Gradio demo for real-time inference on uploaded microscopy images.

## POSITIONS OF RESPONSIBILITY

**TEDxIITHyderabad | Coordinator - Finance & Partnerships**

- Coordinate partnership development and financial planning for TEDxIITHyderabad, spanning sponsor outreach,

partner coordination and event planning, keeping budgets, timelines and partner commitments aligned across the team.
- Organized and hosted the 2026 team orientation, presenting the team's vision, mission and upcoming event roadmap to incoming members and aligning the new team on goals, roles and responsibilities for the year ahead.

**Extra Mural Lectures (EML), IIT Hyderabad | Coordinator - Collaboration & PR**

- Manage collaboration and PR for Extra Mural Lectures, handling external outreach, promotional activities and stakeholder communication to connect speakers, partners and the student community around every EML event.
- Coordinated logistics and crowd management for an EML featuring ISRO astronaut Group Captain Shubhanshu Shukla, delivering a smooth experience for an audience of 700+ students while coordinating with the wider organizing team.

**International Relations Cell, IIT Hyderabad | Volunteer**

- Support international students through campus tours, orientation and engagement activities, helping them settle in and become familiar with IIT Hyderabad's campus, facilities, community and everyday student life.
- Facilitate cross cultural interaction between international and IITH students through organized sports activities.

## RELEVANT COURSES

**Computer Science:** Artificial Intelligence, Machine Learning, Data Structure and Algorithm, Operating System and DBMS.

## SKILLS

**Core CS & HPC:** Python, C++, SQL, DSA, OOP, OS, DBMS, MPI, OpenMP, Git/GitHub

**GenAI & Tools:** RAG, LangChain, LangGraph, ChromaDB, Gemini API, Streamlit, REST APIs

**ML/DL:** NumPy, Pandas, scikit-learn, PyTorch, XGBoost, SHAP

## CERTIFICATIONS

**AI Upskilling Certificate:** Technical Foundations - Qualcomm | AI & ML

**ATF 2025 Stage 2 Candidate:** AlgoUniversity | Python & DSA

**McKinsey.org Forward Program:** McKinsey & Company | Strategy & Leadership

## ACHIEVEMENTS & EXTRACURRICULAR

- Gold Medal, Carrom, Milan, IIT Hyderabad